YAKEEN

NEET 2.0 2027

Physics | Lecture - 01

Units and Measurement

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Todays Goal

Type of P.Q

Todays H/w goal

↳ Vector Assignment-3
Solve Karen hai

→ Basic math → Assignment
no → 6,7,8

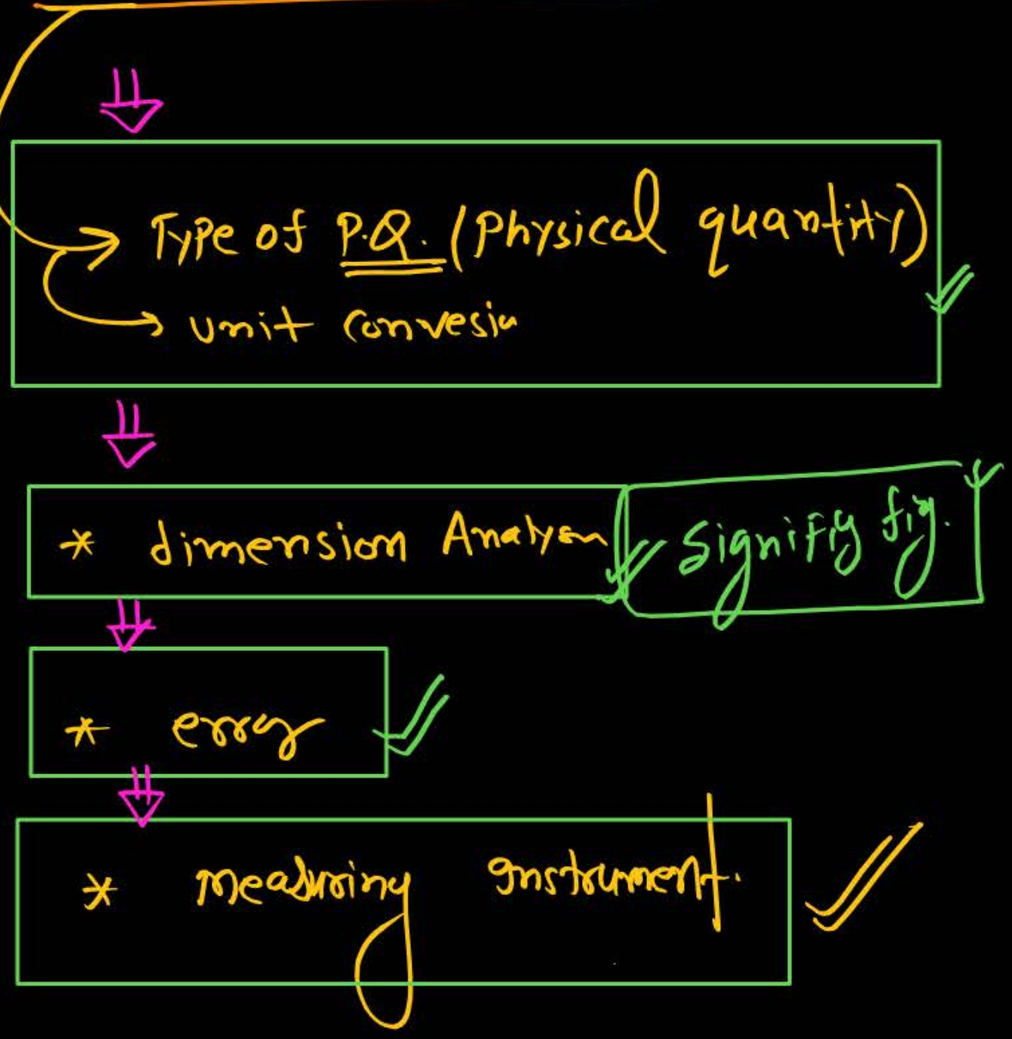
→ Vector Lecture → 7
direction of cosstion

Question Practice advance chahe Ke

(a) Class me Kre ⟶ (11-1.)

(b) assignment me Kre + discuss (ghar pe Toy) (87-1.)

Unit and dimension.



Jab bhi koi aage ke chep of aaye Tension Ka formula Nahi lena, formula rattna nahi hai

Physical Quantity → quantity that can be measured.

Ex – mass, length, time

Fundamental P.Q

→ group of P.Q. which are Independent upon each-other and all derived P.Q. derived from fundument.

1. Mass (kg)
2. length (m)
3. time (s)
4. Tempr (K)
5. Current (A)
6. Amount of sub^{n} (mol)
7. Luminy Intensity (Cd)

Derived P.Q

→ Derived from fundament

- velocity = $\frac{\text{displament}}{\text{time}}$
- Area = $(l)^2$
- density = $\frac{\text{mass}}{(\text{Volume})} = \frac{\text{mass}}{l^3}$

Supplementary P.Q.

Ex – Angle (rad)

Solid Angle (st.rad)

→ have unit but does not have dimension

(1) Which of the following group of Physical quantity may be considered as a group of fundamental P.Q.

MR Scam
✓ (+1)

(a) ✗ mass, Momentum, Velocity ⇒ P = mxv

(b) ✗ force, accn, mass F = mxa.

(c) force, time, Velocity (-1/3)

(d) ✗ Non of these.
70%

fundament P.Q → Independ upon each other
Can't derive from each other

Fundamental P.Q

mass, Length, Time, Temp
kg m s (K) ← fundament unit



(fundament P.Q)

Derived

Derived P.Q

Derived (P.Q)

Derived S.I unit

* $\text{Velocity} = \frac{\text{length}}{\text{time}} = m/sec$

$\text{Speed} = \frac{\text{dist}^n}{\text{time}} = m/s$

* $\text{acceleration} = \frac{\text{Change in Velocity}}{\text{time}} = \frac{\text{length}}{\text{time} \times \text{time}}$

$= \frac{m}{s^2} = m s^{-2}$

* $\text{Force} = \text{mass} \times \text{acin}$

$= kg\ m\ s^{-2}$

* $\text{Momentum} = \text{mass} \times \text{velocit} = kg\ m/sec$

* $\text{Energy} = \frac{1}{2} m (v)^2 = kg \frac{m^2}{s^2}$

$\text{Power} = \frac{\text{energy}}{\text{time}} = \frac{kg\ m^2}{s^2 \times s}$

$= \frac{kg\ m^2}{s^3}$

gf two P.Q have Same unit they they may be same or diff^n P.Q

Girls without make up



Girls after 1 hour make up



Girls after 2 hours



Over all result



Dimension → when any P.Q. express in terms of fundamental P.Q. then Power of fundamental P.Q. is called dimension.

m is not metre

Mass → M^1

length → L^1

time → T^1

Current → A^1

Tempr → θ^1 or K^1

Amount of sub → mol

Lumino Intent → cd

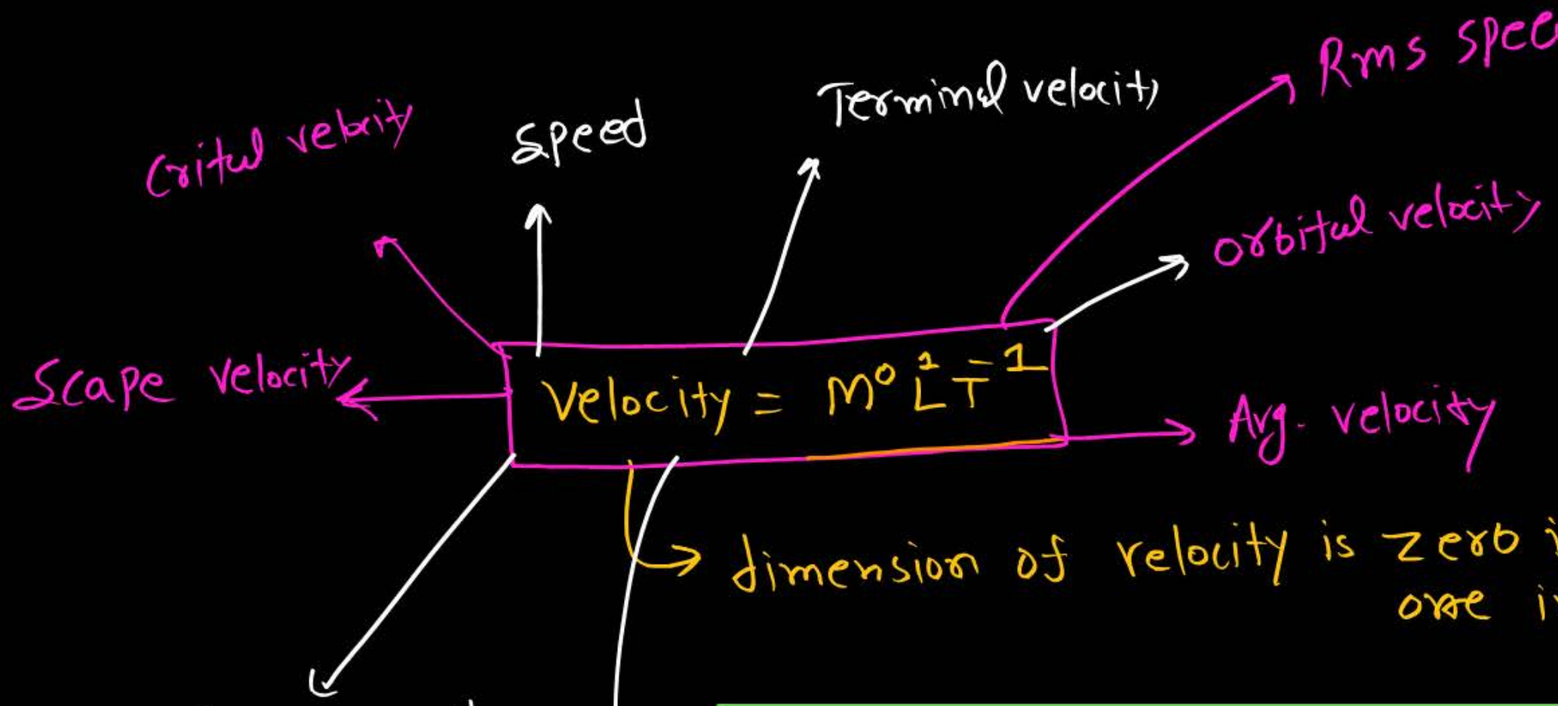
Displament, distance, focal length, Radius of curvature, Height, Range in Projectile, Elongation, Position, wavelength, width, Amplitude, Abc length, circumference, diameter, Mean free Path, fringe width.

→ dimension → $L^1 M^0 T^0$

Question

Which of the following is a characteristic of unit?

1. The unit must be universally accepted
2. It must be invariable and well defined
3. It must be of suitable size and easily available
4. All the above



→ dimension of velocity is zero in mass one in length & -1 in time.

*** MR Sam

→ Angular velocity $(\omega) = \frac{\theta}{T} = M^0 L^0 T^{-1}$

→ $V = R\omega$

$LT^{-1} = L\omega$

$\omega = T^{-1}$ *

⇒ Velocity → LT^{-1}

✓ acceleration = $\frac{\Delta(\text{velocity})}{\text{time}} = \frac{LT^{-1}}{T} = LT^{-2}$

(tangential acc^n)

Force = m x acc^n = $M^1 \times LT^{-2}$

Work = Force x dis^n = $MLT^{-2} \times L$

↳ energy (torque) = ML^2T^{-2}

Power = $\frac{\text{work}}{\text{time}} = \frac{ML^2T^{-2}}{T^1}$

= ML^2T^{-3}

MR Ratta

MR^x Box

Two different P.Q may have Same dimension

(Angle / Angular disp^m) $\theta = M^0L^0T^0$

Angular velocity = $\frac{\theta}{T} = M^0L^0T^{-1}$

Angular acc^n = $\frac{\Delta\omega}{T} = \frac{T^{-1}}{T} = T^{-2}$

$= M^0L^0T^{-2}$

Moment of Inetia $I = mL^2 = M^1L^2$

Angular momentum ⇒ $L = r \times p = L^1(mv)$

$= L^1M^1LT^{-1}$

$= M^1L^2T^{-1}$

Torque = F x length

$= MLT^{-2} \times L$

$= ML^2T^{-2}$ (Same as energy)

Area $= M^0 L^2 T^0$

Volume $= M^0 L^3 T^0$

density $= \frac{\text{mass}}{\text{volume}} = \frac{M^1}{L^3} = ML^{-3}$

Charge $= I \times t = A^1 T^1$

$I = \frac{Q}{t}$

* velocity gradiant $= \frac{\text{velocity}}{\text{length}} = \frac{L T^{-1}}{L} = T^{-1}$

frequency $= \frac{1}{T} = T^{-1}$

* Tempr gradiant $= \frac{\text{Tempr}}{\text{length}} = \frac{K}{L} = K^1 L^{-1}$

frictional force
magnetic force
electric force
Bouyant force
Tension force
Spring force
electrostatic force
Nuclear force
gravitational force

→ MLT^{-2}

electromotive force

→ MLT^{-2}

(a) Yes ✓

(b) NO

Volt (unit) $= \frac{U}{q}$

$= \frac{ML^2T^{-2}}{A^1T^1}$

$= ML^2A^{-1}T^{-3}$

energy $\longrightarrow$ ML^2T^{-2}

K.E

P.E

Heat energy

Light energy

Magnetic energy

Electric energy

Nuclear energy

Work function

Energy density $= \frac{\text{energy}}{\text{Volume}}$

$= \frac{ML^2T^{-2}}{L^3}$

$= ML^{-1}T^{-2}$

DIMENSION LESS PHYSICAL QUANTITY

Copy Dimensionless P.Q

- Angle/ Solid angle/ Strain

  Poisson's ratio, refractive index

  Trigonometry formula/ exponential functions, relative permittivity, efficiency, ratio, pure no. specific gravity

Question

If area of object is $5m^2$ then find its value in CGS unit?

Question

Convert 25 m/sec in CGS unit?

Question

Convert 1 Newton into Dyne.

Question

Convert values of 10 Joule in CGS unit?

Question

If unit of length becomes double than S.I unit of length, then area $5m^2$ in new unit will be?

Question

If unit of length 10m and unit of mass is 5 kg and unit of time is 2 sec then, find value of 10 Joule energy is new system of unit.

Question

If unit of length 10m and unit of mass is 5 kg and unit of time is 2 sec then, find unit of energy in new system.

Question

If unit of length in new system is 2m and unit of mass is 10 kg and unit of time 4 sec then find value of 15 N force in new system.

Question

If unit of length in new system is 2m and unit of mass is 10 kg and unit of time 4 sec then unit of force in new system.

Question

In new system of unit, length is x m and mass is y kg, then value of density 5 kg/m^3 in new system will be?

Question

In new system of unit length is x m and mass is y kg, then unit of density in new system.

Question

Mass of object is m = 10 kg moving with velocity 4 m/s find its momentum in SI and CGS unit.

Question

Mass of object is m = 10 kg moving with velocity 4 m/s find its K.E. in SI and CGS unit.

- Area
- Volume
- Velocity
- Acceleration
- Force
- Energy
- Power
- Momentum
- Impulse

Thank

You